CAMPUS EVENT FINANCIAL PLANNING

Python Programming Case Study

1. Problem Statement

Organizing a campus event requires careful planning of expenses such as venue, equipment, catering, transport, and certificates. The organizers must compare the total event cost with sponsorship and participant contributions to know whether the available income is sufficient. They also need to estimate the minimum number of participants required to cover the remaining cost.

2. Objective

To develop a Python program that calculates the total event cost, estimates total income from sponsorship and participant contributions, finds any financial shortfall, calculates the minimum participants needed to cover the cost after sponsorship, and displays whether the event budget is achieved.

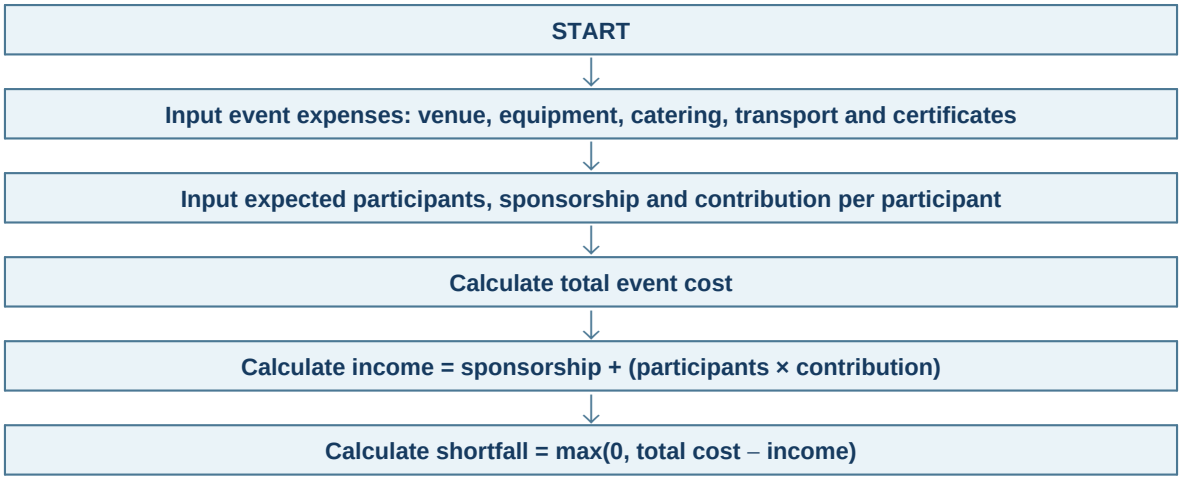
3. Assumptions

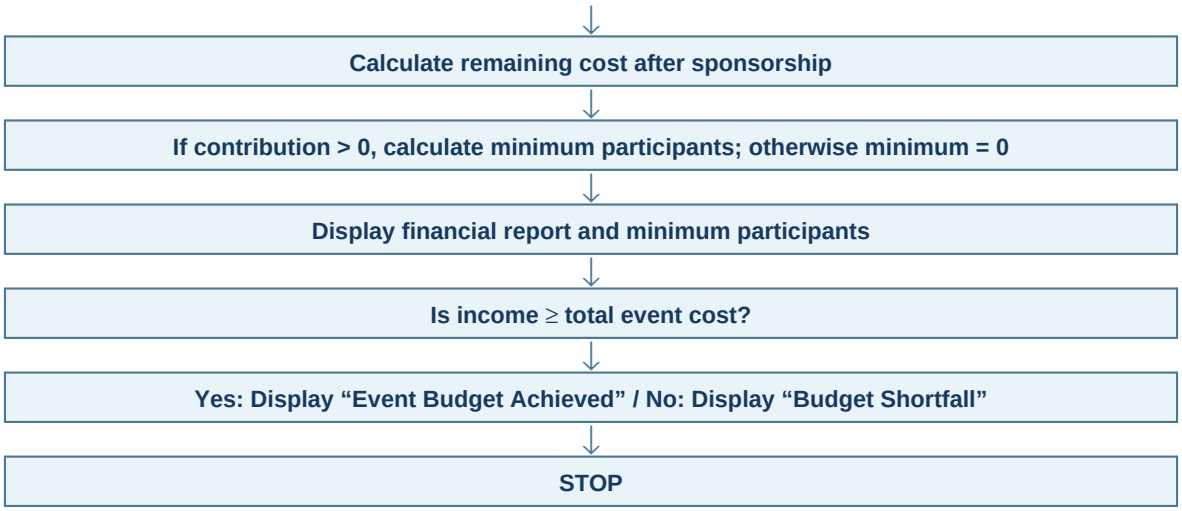
• All costs, sponsorship, and contribution values are entered as valid non-negative numbers.
• The expected participant count is a whole number greater than or equal to zero.
• Total event cost is the sum of venue, equipment, catering, transport, and certificate costs.
• Total income is sponsorship plus expected participants multiplied by contribution per participant.
• Financial shortfall is zero when income covers the cost; otherwise it is the difference between cost and income.
• Minimum participants are calculated using the remaining cost after sponsorship and the contribution per participant. If the contribution is zero, the program reports zero for this calculation.
• The event budget is achieved when total income is greater than or equal to total event cost.

4. Algorithm

1. Start the program.
2. Read venue, equipment, catering, transport, and certificate costs.
3. Read expected participants, sponsorship amount, and contribution per participant.
4. Calculate total event cost by adding all five expense categories.
5. Calculate total income as sponsorship plus participants multiplied by contribution.
6. Calculate financial shortfall as the greater of zero and total cost minus income.
7. Calculate the remaining cost after sponsorship.
8. If contribution per participant is greater than zero, calculate the minimum participants using ceiling division; otherwise set the minimum to zero.
9. Display the total cost, participant contribution total, sponsorship, shortfall, and minimum participants.
10. If income is at least the total cost, display “Event Budget Achieved”; otherwise display “Budget Shortfall”.
11. Stop the program.

5. Flowchart



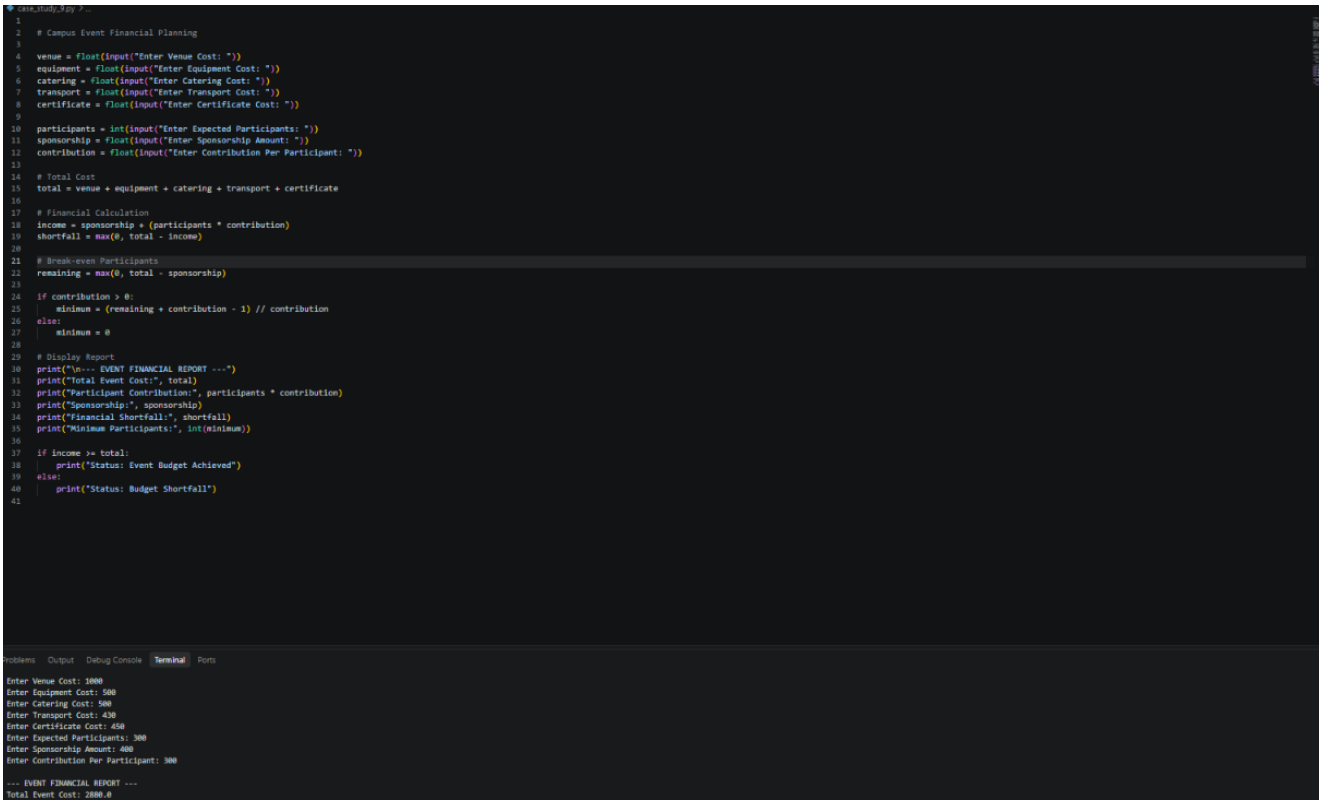

Calculate remaining cost after sponsorship
If contribution > 0, calculate minimum participants; otherwise minimum = 0
Display financial report and minimum participants
Is income ≥ total event cost?
Yes: Display “Event Budget Achieved” / No: Display “Budget Shortfall”
STOP

6. Pseudocode

```
BEGIN
INPUT venue, equipment, catering, transport, certificate
INPUT participants, sponsorship, contribution
total = venue + equipment + catering + transport + certificate
income = sponsorship + (participants * contribution)
shortfall = MAX(0, total - income)
remaining = MAX(0, total - sponsorship)
IF contribution > 0 THEN
minimum = (remaining + contribution - 1) // contribution
ELSE
minimum = 0
END IF
DISPLAY total, participants * contribution, sponsorship
DISPLAY shortfall, minimum
IF income >= total THEN
DISPLAY 'Event Budget Achieved'
ELSE
DISPLAY 'Budget Shortfall'
END IF
END
```

7. Python Source Code and Output Screenshot

The original screenshot supplied for this case study is included below, showing the Python source code and the program output.



8. Conclusion

The Campus Event Financial Planning program helps organizers estimate event expenses, calculate income from sponsorship and participant contributions, identify a possible shortfall, and determine the minimum

participant count needed to cover the cost after sponsorship. It also reports whether the planned income meets the event budget. This case study demonstrates Python input, arithmetic operations, max(), integer division, and conditional statements.